Finding the Missing Factor
Answer Key

Grades 2–3 Operations and Algebraic Thinking

Quick Answers

1. 5	4. 7	7. 5	10. 8, 7
2. 6	5. 7	8. 9	
3. 4	6. 7	9. 9	

Curriculum

Level: Grades 2–3 Operations and Algebraic Thinking
3.OA.A.3 – *problems 6, 7, 10*
3.OA.C.7 – *problems 1, 2, 3, 4, 5, 8, 9*
Difficulty 1–4 of 5 across 11 tagged checks.

Common wrong answers

5. *If they got 54: took the numbers apart instead of dividing, writing 63 minus 9*
8. *If they got 64: subtracted the two numbers in the sentence instead of dividing*
9. *If they got 32: subtracted 4 from 36 instead of finding the missing factor*

1. Array of tiles: $4 \times \square = 20$.

Step 1. The rows are equal, so the picture is a multiplication situation: number of rows times tiles in each row equals the total.
Step 2. The missing number is the one that turns 4 groups into 20, which is the division fact $20 \div 4$.
Step 3. $20 \div 4 = 5$, and the check multiplies back: $4 \times 5 = 20$. Each row holds $\boxed{5}$
Listen for: a child who counts every tile one at a time. Ask them to count one row and then skip-count by that number.

2. Three equal bags: $3 \times \square = 18$.

Step 1. Equal bags means equal groups, so the missing factor is how many marbles sit in one bag.
Step 2. Use the matching division fact: $18 \div 3$.
Step 3. $18 \div 3 = 6$, and $3 \times 6 = 18$ checks it. One bag holds $\boxed{6}$
Teaching note: the picture shows the same six dots three times — point at one box and say “this is one group” before counting.

3. Seating chart: $\square \times 7 = 28$.
Step 1. Here the size of each group is known (7 seats in a row) and the number of groups is missing, so the missing factor counts the rows.
Step 2. Divide the total by the size of one group: $28 \div 7$.
Step 3. $28 \div 7 = 4$, and $4 \times 7 = 28$ checks it. The hall has $\boxed{4}$ rows.
Teaching note: this is the same fact family as problem 3 read the other way. Say it aloud both ways: "four sevens are twenty-eight".

4. Number sentence: $\square \times 8 = 56$.
Step 1. No picture here, so lean on the fact family: the missing factor, 8, and 56 all belong to one family.
Step 2. The division fact that names the missing number is $56 \div 8$.
Step 3. $56 \div 8 = 7$, and multiplying back gives $7 \times 8 = 56$. The missing factor is $\boxed{7}$
Listen for: guess-and-check that stops at the first product under the total. The check step catches it.

5. Number sentence: $9 \times \square = 63$.
Step 1. The problem asks for the division fact on purpose, because every missing-factor question has one waiting for it.
Step 2. The total is 63 and one factor is 9, so the division fact is $63 \div 9$.
Step 3. $63 \div 9 = 7$, and $9 \times 7 = 63$ checks it. The missing factor is $\boxed{7}$
Common wrong answer: 54 — the two numbers were subtracted. Subtraction takes an amount away; it does not split a total into equal groups.

6. 42 **chairs in** 6 **equal rows.**
Step 1. "Equal rows" names the groups, so the story is $6 \times (\text{chairs in a row}) = 42$.
Step 2. The missing factor is $42 \div 6$.
Step 3. $42 \div 6 = 7$, and $6 \times 7 = 42$ checks it. Each row holds $\boxed{\text{7 chairs}}$
Teaching note: the unit belongs in the answer. "Seven" alone does not say seven of what.

7. 45 **stickers packed** 9 **to a pack.**
Step 1. This time the size of each group is given and the number of groups is missing, so the equation is $9 \times (\text{number of packs}) = 45$.
Step 2. The missing factor is $45 \div 9$.
Step 3. $45 \div 9 = 5$, and $9 \times 5 = 45$ checks it. She fills $\boxed{\text{5 packs}}$
Listen for: an answer of 9. That is the number already given — ask "how many packs, not how many in a pack?"

8. Division sentence: $72 \div \square = 8$.
Step 1. A division sentence with the divisor missing is still a missing-factor question: it says some number of groups of 8 makes 72.
Step 2. Rewrite it as $\square \times 8 = 72$, so the missing number is $72 \div 8$.
Step 3. $72 \div 8 = 9$, and the multiplication check is $9 \times 8 = 72$. The missing number is $\boxed{9}$
Common wrong answer: 64 — the numbers in the sentence were subtracted rather than divided.

9. Error analysis: Mia wrote 32 for $4 \times \square = 36$.

Step 1. Test Mia's number in her own sentence: 4×32 is far larger than 36, so her answer cannot be right.
Step 2. What Mia did was subtract: $36 - 4 = 32$. A missing factor is found by splitting the total into equal groups, not by taking one number away from the other.
Step 3. The correct missing factor is $36 \div 4$, and $36 \div 4 = 9$. The check is $4 \times 9 = 36$, so the box holds $\boxed{9}$
What to say: "Four groups of thirty-two would be enormous. Does your answer make the sentence true?"

10. 56 counters in equal rows.

Step 1. Both parts describe the same 56 counters shared into equal rows, so each part is one missing-factor equation.
Step 2. Part (a): $7 \times \square = 56$, so the missing factor is $56 \div 7$. That gives 8 counters in each row.
Step 3. Part (b): $8 \times \square = 56$, so the missing factor is $56 \div 8$. That gives 7 counters in each row.
Part (a) answer: $\boxed{8}$
Part (b) answer: $\boxed{7}$
Why it flips: the total never changed, so more rows must mean fewer counters in each row — the two answers are the same fact family read both ways.